

SimplePlus Beamer Theme

Subtitle

Zhenxiao Chen

Department of Economics
University of Pennsylvania

May 7, 2026

Overview

1. Problem Setup

2. Motivation

Preview

- A simple statistical treatment allocation problem

$$\max_{\phi \in \Phi} \mathbb{E}[Y(1)\phi(X) + Y(0)(1 - \phi(X))]$$

- The first-best:

$$\phi^*(x) = \mathbb{1}\{h_0(x) \geq 0\}, \quad h_0(x) := \mathbb{E}[Y(1) - Y(0) \mid X = x] = \text{CATE}(x)$$

- Policy learning: What is the best policy under a (un)constrained policy class

$$\hat{\phi}(x) = \mathbb{1}\{x \in G^*\} \in \arg \min_{G \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i D_i}{e(X_i)} - \frac{Y_i(1 - D_i)}{1 - e(X_i)} \right) \mathbb{1}\{X_i \in G\}$$

- **Inference on optimal policy value:** Draw inference on the value of the first-best rule

$$W(h_0) = \int_{\mathcal{X}} \mathbb{1}\{h_0(x) \geq 0\} h_0(x) f(x) dx, \quad V(h_0) = \int_{\mathcal{X}} \mathbb{1}\{h_0(x) \geq 0\} v_0(x) f(x) dx$$

Preview

- In ?, we studied an upper-contour integral functional

$$\Phi(h_0) = \int_{\mathcal{X}} \mathbb{1}\{h_0(x) \geq 0\} \phi(h_0(x), x) f(x) dx = \int_{\{h_0(x) \geq 0\}} \phi(h_0(x), x) f(x) dx,$$

where the boundary $\{h_0(x) = 0\}$ is smooth and a well-defined submanifold.

- $\Phi(h_0) = W(h_0)$ when $\phi(h_0(x), x) = h_0(x)$
- $\Phi(h_0) = V(h_0)$ when $\phi(h_0(x), x) = v_0(x)$

Preview

- Contribution: Compute the pathwise derivative via the [Generalized Leibniz Rule](#)

$$\left. \frac{d\Phi(h_0 + t(h - h_0))}{dt} \right|_{t=0} = \int \mathbb{1}\{h_0(x) \geq 0\} \partial_1 \phi(h_0(x), x) (h(x) - h_0(x)) f(x) dx \\ + \int_{\{x \in \mathcal{X} : h_0(x) = 0\}} \frac{h(x) - h_0(x)}{\|\nabla_x h_0(x)\|} \phi(h_0(x), x) f(x) d\mathcal{H}^{d-1}(x)$$

- Pathwise derivative captures the first-order impact of estimating h_0 for $W(h_0)$
- The first-order impact is two-fold: one comes from perturbing the integrand; the other comes from perturbing the boundary $\{x : h_0(x) = 0\}$
- [The second term is nonstandard](#): does not have a well-defined Riesz representer in $L_2(P) \Rightarrow$ plug-in estimator does not converge in parametric rate

- Contribution: Inference for regular and irregular optimal policy values

$$\frac{\sqrt{n}(\Phi(\hat{h}) - \Phi(h_0))}{V_K^{1/2}} \xrightarrow{d} \mathcal{N}(0, 1), \quad V_K = \mathbb{E}[(\nu_K^*(X_i)\epsilon_i)^2]$$

- Consider a linear sieve semiparametric plug-in estimator $\Phi(\hat{h})$ for $\Phi(h_0)$
- Approximate $L_2(P)$ using a sequence of linear sieve spaces, where sieve Riesz representer is guaranteed to exist
- It is a **unifying procedure** for both cases

My Goal

- I wanna take a step further to study a **double(multi)-threshold functional** in the context of a statistical treatment allocation problem

$$\int_{\mathcal{X}} \mathbb{1}\{h_0(x) \geq 0\} \mathbb{1}\{g_0(x) \geq 0\} \phi(h_0(x), g_0(x), x) f(x) dx$$

- It it is a natural but non-trivial extension to ?
- It emerge naturally in many empirically relevant contexts $\implies$ appealing to practitioners

Comparison with ?

Double-threshold functional:

$$\int_{\mathcal{X}} \mathbb{1}\{h_0(x) \geq 0\} \mathbb{1}\{g_0(x) \geq 0\} \phi(h_0(x), g_0(x), x) f(x) dx$$

- The boundary $\{\min\{h_0(x), g_0(x)\} = 0\}$ is not smooth but is only **piece-wise smooth**
 $\implies ?$ is not directly applicable
- Two natural ways to proceed:
 - (1) Smooth min with softmin and try to go back to ?; however, smoothing introduces approximation bias
 - (2) Adapt the Generalized Leibniz Rule to accommodate a piece-wise smooth boundary; perturbing each boundary while holding the rest fixed
- Overall, I want to extend techniques in ? to more complicated settings (allocation problem or statistics)

Motivations

Double-threshold functionals arise naturally in treatment allocation problems where:

- strict overlap is imposed in observational studies for statistical stability;
- the policymaker faces an admissibility-constrained treatment-choice problem;
- the target is a summary statistic of the optimal policy in a data environment with richer causal structure;
- the allocation problem is more complex, as in dynamic treatment regimes.

In the discussion of motivations, I will also compare my paper with the key references in the existing literature.

Strict Overlap in Observational Data

- **Weak overlap**, together with unconfoundedness, point-identifies $h_0(x)$ and thus $W(h_0)$

$$\begin{aligned} & 0 < p_0(D = 1 \mid X = x) < 1 \text{ and } (Y(1), Y(0)) \perp D \mid X \\ \Rightarrow h_0(X) &:= \mathbb{E}[Y(1) - Y(0) \mid X] = \mathbb{E}[Y \mid X, D = 1] - \mathbb{E}[Y \mid X, D = 0] \\ \Rightarrow W(h_0) &= \int_{\mathcal{X}} \mathbb{1}\{h_0(x) \geq 0\} h_0(x) f(x) dx \end{aligned}$$

- In practice, however, researchers often impose **strict overlap**, a stronger condition used for stable estimation and inference:

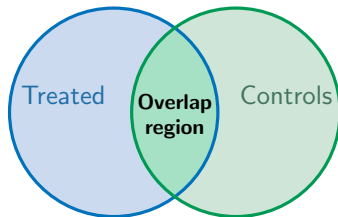
$$c < p_0(D = 1 \mid X = x) < 1 - c, \quad \text{for some constant } c \in (0, 1/2).$$

Example 1: Strict Overlap in Observational Data

- Strict overlap avoids extreme propensity scores, which imply limited data on either treated or control units in some regions of the support and lead to unstable estimates:

$$\hat{h}(x) = \hat{\mathbb{E}}[Y \mid X = x, D = 1] \\ - \hat{\mathbb{E}}[Y \mid X = x, D = 0], \quad W(\hat{h}).$$

- Sensible CATE estimates come from covariate regions where both groups are observed.



where comparisons are statistically supported

Strict Overlap in Observational Data

- Strict overlap is typically guaranteed by design in many RCTs, but it is **not automatic in observational data** $\implies$ we restrict attention to regions with sufficient overlap
- This changes the welfare object from $W(h_0)$ to an overlap-restricted criterion:

$$W_\alpha(h_0) = \int_{\mathcal{X}} \mathbb{1}\{h_0(x) \geq 0\} \mathbb{1}\{\alpha < p_0(D = 1 \mid X = x) < 1 - \alpha\} h_0(x) f(x) dx$$

- This also aligns with the common practice of trimming observations with extreme estimated propensity scores:

$$\alpha \equiv 0.05, 0.1, \dots$$

Admissibility-Constrained Treatment-Choice Problem

- A policymaker often faces a constrained treatment-allocation problem (?):

$$\max_{\phi \in \Phi} \mathbb{E}[Y_1\phi(X) + Y_0(1 - \phi(X))] \quad \text{s.t.} \quad \mathbb{E}[Z_1\phi(X) + Z_0(1 - \phi(X))] \leq \alpha.$$

- Y_1 and Y_0 are welfare outcomes with and without treatment;
 - Z_1 and Z_0 are the corresponding costs under treatment and control.
- The optimal rule treats covariate type $X = x$ when the welfare gain is large enough to justify the additional cost:

$$\mathbb{1} \{ \mathbb{E}[Y_1 - Y_0 \mid X = x] - k\mathbb{E}[Z_1 - Z_0 \mid X = x] \geq 0 \}.$$

- k is the shadow price of the resource constraint;
- Smaller $\alpha \Rightarrow$ larger k ; larger $\alpha \Rightarrow$ smaller k ;
- This criterion is **compensatory**: sufficiently large benefits can justify higher costs.

Admissibility-Constrained Treatment-Choice Problem

- In some settings, however, a high cost cannot be compensated by a high benefit:
 - Example: severe side effects, measured by $\mathbb{E}[Z_1 - Z_0 \mid X]$.
 - If the **side effect** is sufficiently severe, the policymaker may refuse treatment regardless of the expected benefit.
- This corresponds to a **checklist-style admissibility criterion**:
 - Treatment is considered only for individuals who pass the safety screen;
 - No health benefit can compensate for failing the safety screen;
 - This echoes the Hippocratic principle: **"First, do no harm."**

Admissibility-Constrained Treatment-Choice Problem

- The policymaker's problem becomes:

$$\max_{\phi \in \Phi} \mathbb{E}[Y_1 \phi(X) + Y_0(1 - \phi(X))]$$

$$\text{s.t. } \phi(x) \leq \mathbb{1}\{\mathbb{E}[Z_1 - Z_0 \mid X = x] \leq \bar{z}\}, \quad \forall x \in \mathcal{X}.$$

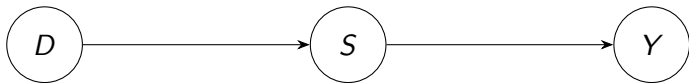
- The optimal welfare under this treatment rule is:

$$\int_{\mathcal{X}} \mathbb{1}\{\mathbb{E}[Y_1 - Y_0 \mid X = x] \geq 0\} \mathbb{1}\{\mathbb{E}[Z_1 - Z_0 \mid X = x] \leq \bar{z}\} \mathbb{E}[Y_1 - Y_0 \mid X = x] f(x) dx.$$

Descriptive Statistics

- To better understand the consequences of implementing an optimal policy, policymakers may want summary statistics beyond the welfare value:
 - average characteristics of those assigned to treatment;
 - average cost of the policy;
 - population share assigned to treatment.
- With richer data and more structured causal models, these summary statistics can involve additional causal objects.
- This naturally leads to double-threshold functionals: one threshold defines the treatment rule, while the other describes a policy-relevant feature of the population.

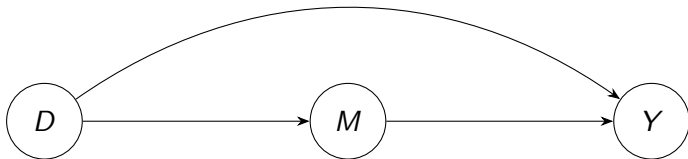
Descriptive Statistics: Surrogate Outcomes



- **Surrogate index setting (?)**:
 - experimental data identify the effect of D on short-term surrogates S ;
 - auxiliary observational data link surrogates S to long-term outcome Y .
- A policymaker may assign treatment based on short-term surrogate gains, but still want to know whether long-run gains are positive.
- A useful summary statistic is the population share treated by the surrogate-based rule whose long-run treatment effect is also positive:

$$\int_{\mathcal{X}} \mathbb{1}\{\mathbb{E}[S(1) - S(0) \mid X = x] \geq 0\} \mathbb{1}\{\mathbb{E}[Y(1) - Y(0) \mid X = x] \geq 0\} f(x) dx$$

Descriptive Statistics: Mediation



- **Causal mediation analysis** (?) studies how treatment effects operate through intermediate mechanisms.
- Example: D is access to cognitive behavioral therapy, M is household support, and Y is financial empowerment.
- A useful summary statistic is the population share optimally treated for Y whose mediated contribution through M is positive:

$$\int_{\mathcal{X}} \mathbb{1}\{\mathbb{E}[Y(1) - Y(0) \mid X = x] \geq 0\} \mathbb{1}\{\mathbb{E}[Y(1, M(1)) - Y(1, M(0)) \mid X = x] \geq 0\} f(x) dx$$

Dynamic Treatment Regime

- Consider the setting in ? and ?: We assume access to i.i.d. samples $Z \sim P_Z$ of the form $Z = (X_1, A_1, X_2, A_2, Y)$ that are generated according to the following structural equation model.

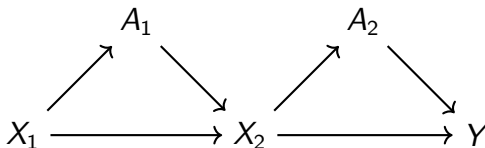
$$Y = f_Y(A_2, X_2, \xi_Y)$$

$$A_2 = f_{A_2}(X_2, \xi_{A_2})$$

$$X_2 = f_{X_2}(A_2, X_1, \xi_{X_2})$$

$$A_1 = f_{A_1}(X_1, \xi_{A_1})$$

$$X_1 = f_{X_1}(\xi_{X_1})$$



Dynamic Treatment Regime

- Dynamic treatment regime:

$$\pi_1 : \mathcal{X}_1 \rightarrow \{0, 1\}$$

$$\pi_2 : \mathcal{X}_2 \rightarrow \{0, 1\}$$

- Potential outcome under dynamic treatment regime

$$Y := Y^{(A_1, A_2)}$$

$$Y^{(\tau_1, \pi_2)} := \sum_{\tau_2 \in \{0, 1\}} \mathbb{1} \left\{ \pi_2 \left(X_2^{(\tau_1)} \right) = \tau_2 \right\} Y^{(\tau_1, \tau_2)}.$$

$$Y^{(\pi_1, \pi_2)} := \sum_{\tau_1 \in \{0, 1\}} \mathbb{1} \{ \pi_1(X_1) = \tau_1 \} Y^{(\tau_1, \pi_2)}.$$

Dynamic Treatment Regime

- Assume sequential independence and perform backward induction to solve for the optimal dynamic treatment regime

$$Y^{(T_1, \pi_2)} \perp A_2 \mid X_2, \quad Y^{(\pi_1, \pi_2)} \perp A_1 \mid X_1$$

- Optimal treatment in period 2:

$$\pi_2^*(x_2) \in \arg \max_{a \in \{0,1\}} \mu_a(x_2), \quad \mu_a(x) = \mathbb{E}[Y \mid X_2 = x_2, A_2 = a]$$

- Optimal treatment in period 1:

$$\pi_1^*(s) \in \arg \max_{\tau_1 \in \{0,1\}} \mathbb{E}[V_2(X_2) \mid X_1 = x_1, A_1 = \tau_1], \quad V_2(x_2) := \max\{\mu_0(x_2), \mu_1(x_2)\}$$

Dynamic Treatment Regime

- Total optimal policy value:

$$\begin{aligned} V^* &= \mathbb{E} \left[Y^{(\pi_1^*, \pi_2^*)} \right] \\ &= \mathbb{E} \left[Y^{(0,0)} \mathbb{1}_{\{\pi_1^*(X_1) = 0, \pi_2^*(X_2^{(0)}) = 0\}} \right] + \mathbb{E} \left[Y^{(1,0)} \mathbb{1}_{\{\pi_1^*(X_1) = 1, \pi_2^*(X_2^{(1)}) = 0\}} \right] \\ &\quad + \mathbb{E} \left[Y^{(0,1)} \mathbb{1}_{\{\pi_1^*(X_1) = 0, \pi_2^*(X_2^{(0)}) = 1\}} \right] + \mathbb{E} \left[Y^{(1,1)} \mathbb{1}_{\{\pi_1^*(X_1) = 1, \pi_2^*(X_2^{(1)}) = 1\}} \right] \end{aligned}$$

- Using a smoothing technique, ? shows V^* is parametrically estimable and constructs corresponding CIs.
- However, it might be interesting to draw inference on **welfare coming from each treatment path**, say

$$V_{11}^* = \mathbb{E} \left[Y^{(1,1)} \mathbb{1}_{\{\pi_1^*(X_1) = 1, \pi_2^*(X_2^{(1)}) = 1\}} \right]$$

Dynamic Treatment Regime

$$\begin{aligned} V_{11}^* &= \mathbb{E} \left[Y^{(1,1)} \mathbb{1} \left\{ \pi_1^*(X_1) = 1, \pi_2^*(X_2^{(1)}) = 1 \right\} \right] \\ &= \int_{\mathcal{X}_2} \int_{\mathcal{X}_1} \mathbb{1} \{ \delta(x_2) \geq 0 \} \mathbb{1} \{ \kappa(x_1) \geq 0 \} \mu_1(x_2) p_1(x_2 \mid x_1) m(x_1) dx_1 dx_2 \end{aligned}$$

with

$$\delta(x_2) = \mu_1(x_2) - \mu_0(x_2)$$

$$\kappa(x_1) = \mathbb{E}[V_2(X_2) \mid X_1 = x_1, A_1 = 1] - \mathbb{E}[V_2(X_2) \mid X_1 = x_1, A_1 = 0]$$

- The boundary $\{\kappa(x_1) = 0\}$ is smooth because max is smoothed out by conditional expectation

Dynamic Treatment Regime

- Assuming $p_1(x_2 \mid x_1) = \Pr(X_2 = x_2 \mid X_1 = x_1, A_1 = 1)$ is known, I have shown that V_{11}^* is irregular
- In particular, $DV_{11}^*[\dot{\mu}_0, \dot{\mu}_1] = \text{standard linear functional} + B_{11}^{M_1} + B_{11}^{M_2}$
- Why is V^* regular? This is because boundary terms cancel out.

$$B_{11}^{M_1} + B_{10}^{M_1} + B_{01}^{M_1} + B_{00}^{M_1} = 0$$

$$B_{11}^{M_2} + B_{10}^{M_2} + B_{01}^{M_2} + B_{00}^{M_2} = 0$$

Conclusion

In this paper,

- I want to study inference on optimal policy values or related statistics in settings existing work could not handle;
- I would like to extend the advanced machinery in ? to these new settings;
- I would hope to sell this idea to practitioners.

References

The End